CMSC 137

Data Communications and Networking

Lecture Module
20 – IP Support Protocols

University of the Philippines
LOS BAÑOS

Module 20: IP Support Protocols

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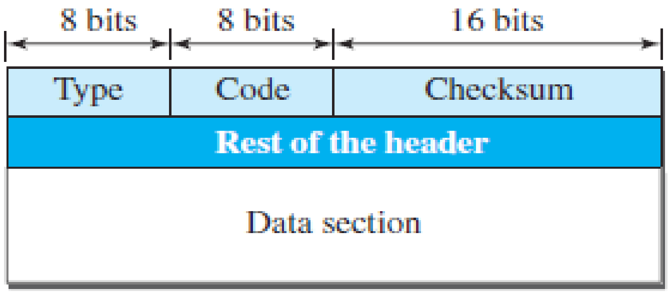
Support Protocols

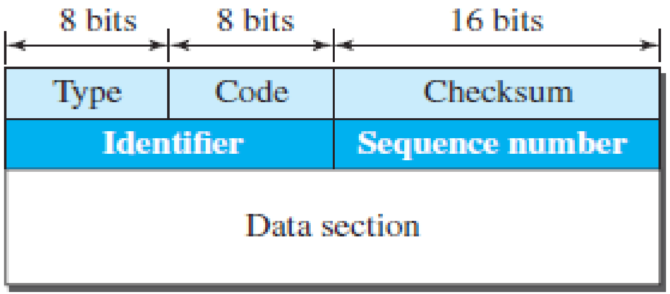
ICMPv4

- IPv4 has deficiencies
- No error-reporting or error correcting mechanism
- Lacks mechanism for host and management queries
- Internet Control Message Protocol version 4 (ICMPv4) compensate for the deficiencies, companion protocol to IPv4 [RFC 792]
- Error-reporting messages – report problems that a router or a host may encounter when it process an IP packet
- Query messages – helps a host get specific information from a router or another host

ICMPv4 Format



Error-reporting messages



Query messages

Type and code values

Error-reporting messages

03: Destination unreachable (codes 0 to 15)
04: Source quench (only code 0)
05: Redirection (codes 0 to 3)
11: Time exceeded (codes 0 and 1)
12: Parameter problem (codes 0 and 1)

Query messages

08 and 00: Echo request and reply (only code 0)
13 and 14: Timestamp request and reply (only code 0)

Debugging Tools

- ping
  - To find if a host is alive and responding
  - Can calculate round-trip time

```
D:\>ping www.google.com -t

Pinging www.google.com [172.217.24.36] with 32 bytes of data:
Reply from 172.217.24.36: bytes=32 time=20ms TTL=53
Reply from 172.217.24.36: bytes=32 time=21ms TTL=53
Reply from 172.217.24.36: bytes=32 time=21ms TTL=53
Reply from 172.217.24.36: bytes=32 time=21ms TTL=53
Reply from 172.217.24.36: bytes=32 time=21ms TTL=53
Reply from 172.217.24.36: bytes=32 time=20ms TTL=53
Reply from 172.217.24.36: bytes=32 time=21ms TTL=53

Ping statistics for 172.217.24.36:
    Packets: Sent = 7, Received = 7, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 20ms, Maximum = 21ms, Average = 20ms
```

Debugging Tools

- traceroute (UNIX) / tracert (Windows)
  - Used to trace path of a packet from a source to the destination
  - Can find all the IP addresses of all routers that are visited along the path (usually set to 30 hops max)

```
D:\>tracert www.google.com

Tracing route to www.google.com [172.217.24.36]
over a maximum of 30 hops:

  1    <1 ms     1 ms     1 ms  10.0.100.254
  2     1 ms     1 ms     1 ms  10.255.255.254
  3     1 ms     1 ms     1 ms  202.92.144.254
  4     1 ms     1 ms     1 ms  ge-0.626-802.1q-vlan-subif.core-7304-irri.pregi.net [202.90.129.237]
  5     6 ms     5 ms     5 ms  ge-1-3.border-asti.pregi.net [202.90.129.253]
  6     6 ms     6 ms     6 ms  tengige0-0-2-0.border-asti-asr.pregi.net [202.90.132.229]
  7    21 ms    20 ms    21 ms  tengige0-0-2-0.border-hk-asr.pregi.net [202.90.129.50]
  8    20 ms    21 ms    20 ms  72.14.212.20
  9    21 ms    21 ms    21 ms  108.170.241.1
 10    21 ms    20 ms    20 ms  108.170.238.131
 11    20 ms    20 ms    21 ms  hkg07s23-in-f36.1e100.net [172.217.24.36]

Trace complete.
```

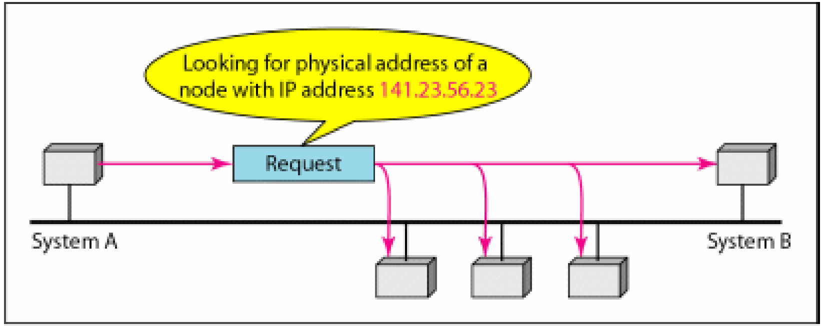
Address Mapping

Address Mapping

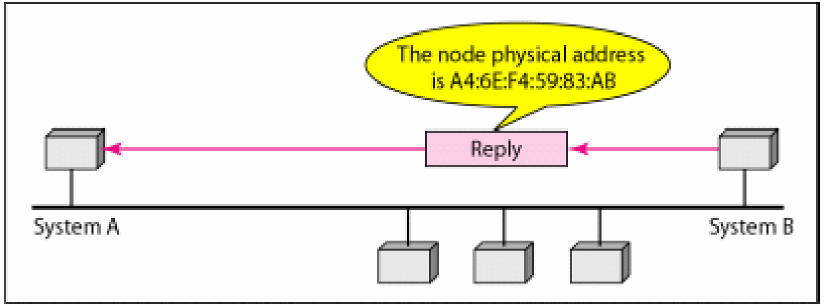
- Physical addresses and logical addresses are two different identifiers and are needed both:
  - A physical network such as Ethernet can have 2 different protocols at the network layer such as IP and IPX (Novell) at the same time
  - A packet at a network layer such as IP may pass through different physical networks such as Ethernet and LocalTalk (Apple)

Mapping Logical to Physical Address

Address Resolution Protocol (ARP)

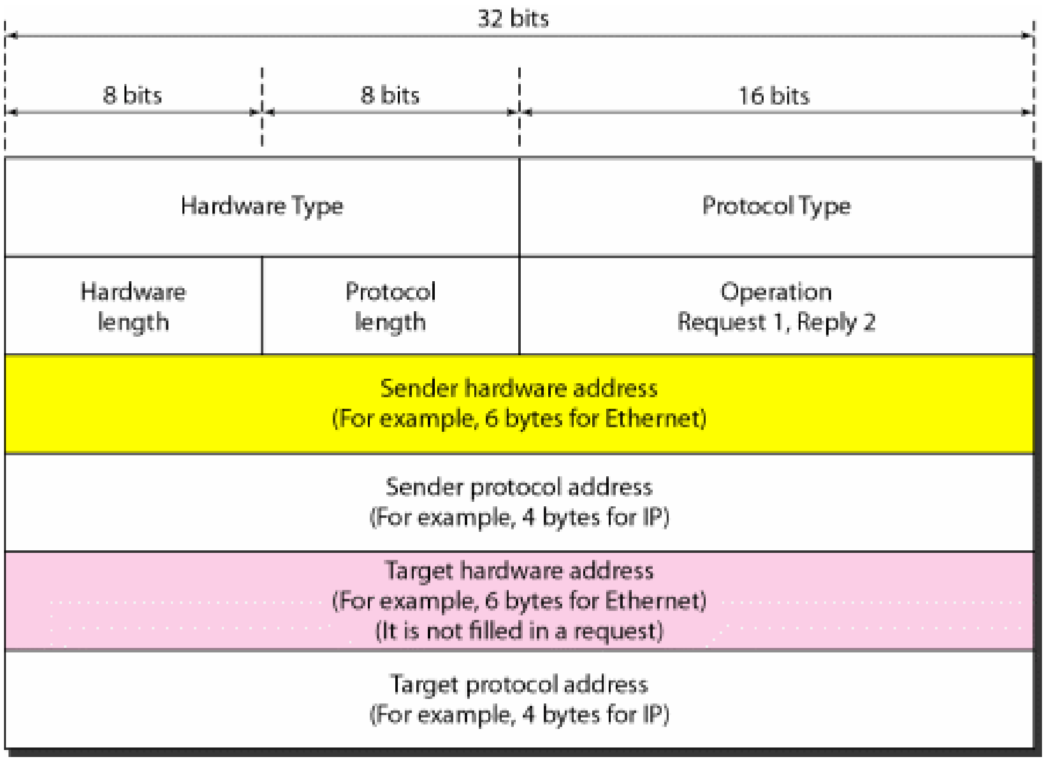


a. ARP request is broadcast



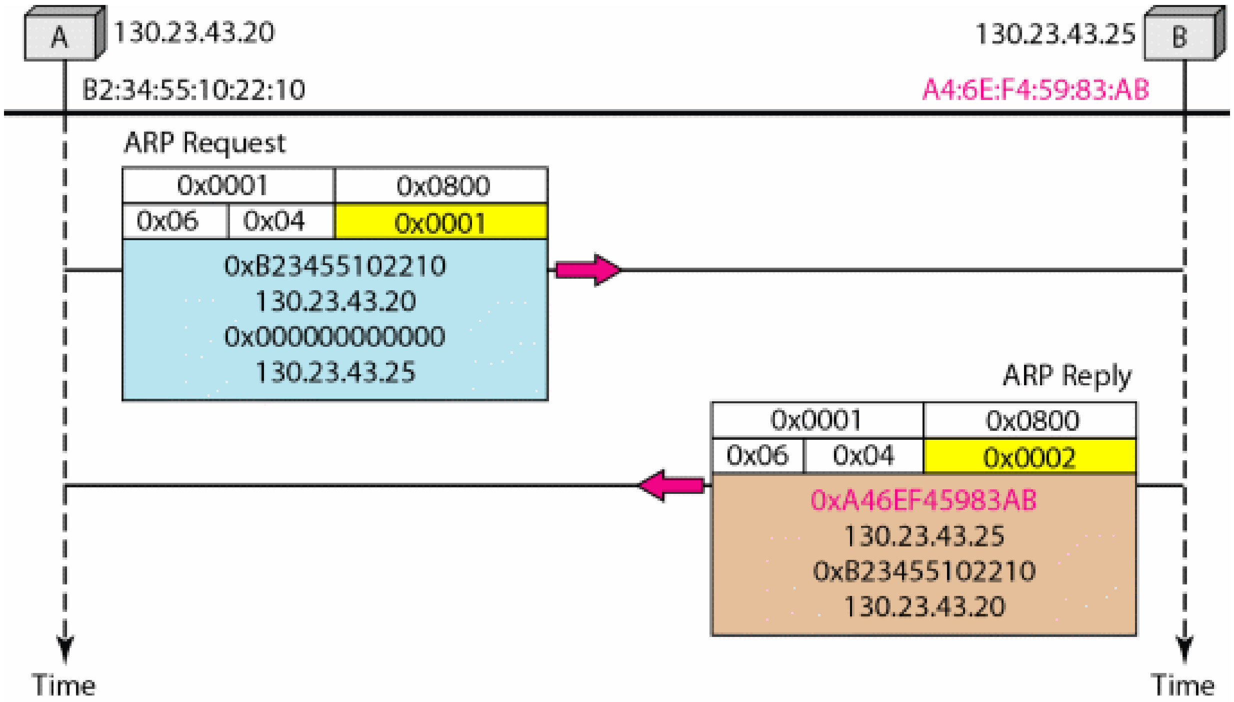
b. ARP reply is unicast

ARP Packet



32 bits
8 bits
8 bits
16 bits
Hardware Type
Protocol Type
Hardware length
Protocol length
Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)
Sender protocol address (For example, 4 bytes for IP)
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)
Target protocol address (For example, 4 bytes for IP)

ARP


A
130.23.43.20
B2:34:55:10:22:10
130.23.43.25
B
A4:6E:F4:59:83:AB
ARP Request
0x0001
0x0800
0x06
0x04
0x0001
0xB23455102210
130.23.43.20
0x000000000000
130.23.43.25
ARP Reply
0x0001
0x0800
0x06
0x04
0x0002
0xA46EF45983AB
130.23.43.25
0xB23455102210
130.23.43.20
Time
Time

Mapping Physical to Logical Address

- Occasions in which host know its physical address but needs to know its logical address
  - A diskless station is booted
  - An organization does not have enough IP addresses to assign to each station
- RARP, BOOTP and DHCP

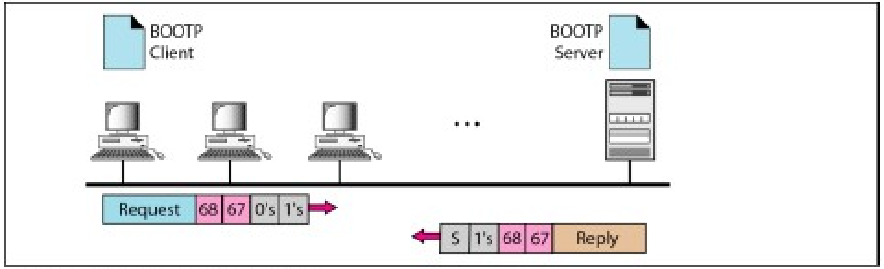
RARP

- Reverse Address Resolution Protocol
  - A RARP request is created and broadcast on the local network
  - Another machine on the local network that knows all the IP address will respond to the RARP reply
  - Problem: broadcasting is done at the data link layer, it does not pass the boundaries of a network

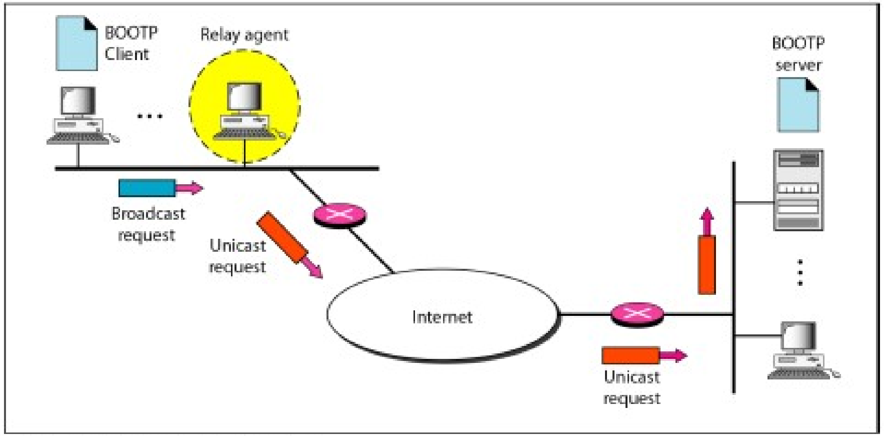
BOOTP

Bootstrap Protocol – application layer protocol used to automatically assign an IP address to network devices from a configuration server

[RFC 951]



a. Client and server on the same network

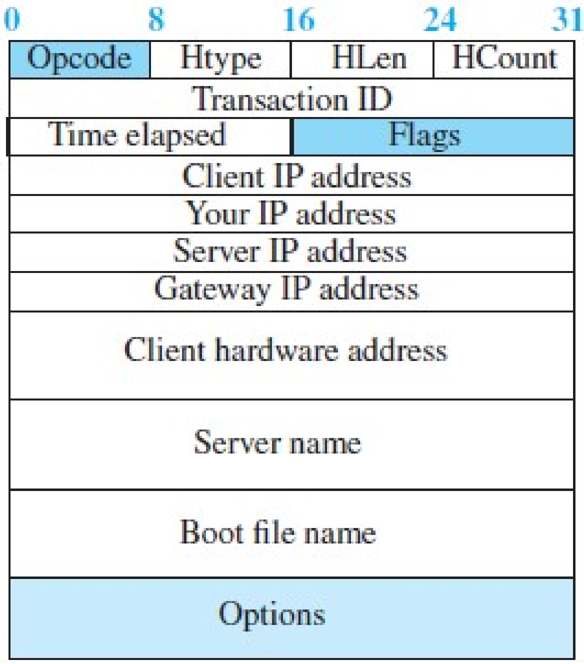


b. Client and server on different networks

DHCP

- Dynamic Host Configuration Protocol – provides static and dynamic addresses allocation that can be manual or automatic
- Static address allocation – acts as what BOOTP does. DHCP server has a databases that statically binds physical addresses to IP addresses
- Dynamic address allocation – DHCP has a second database with pool of available IP addresses

DHCP Message Format

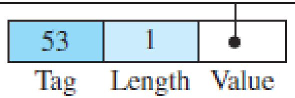


Fields:

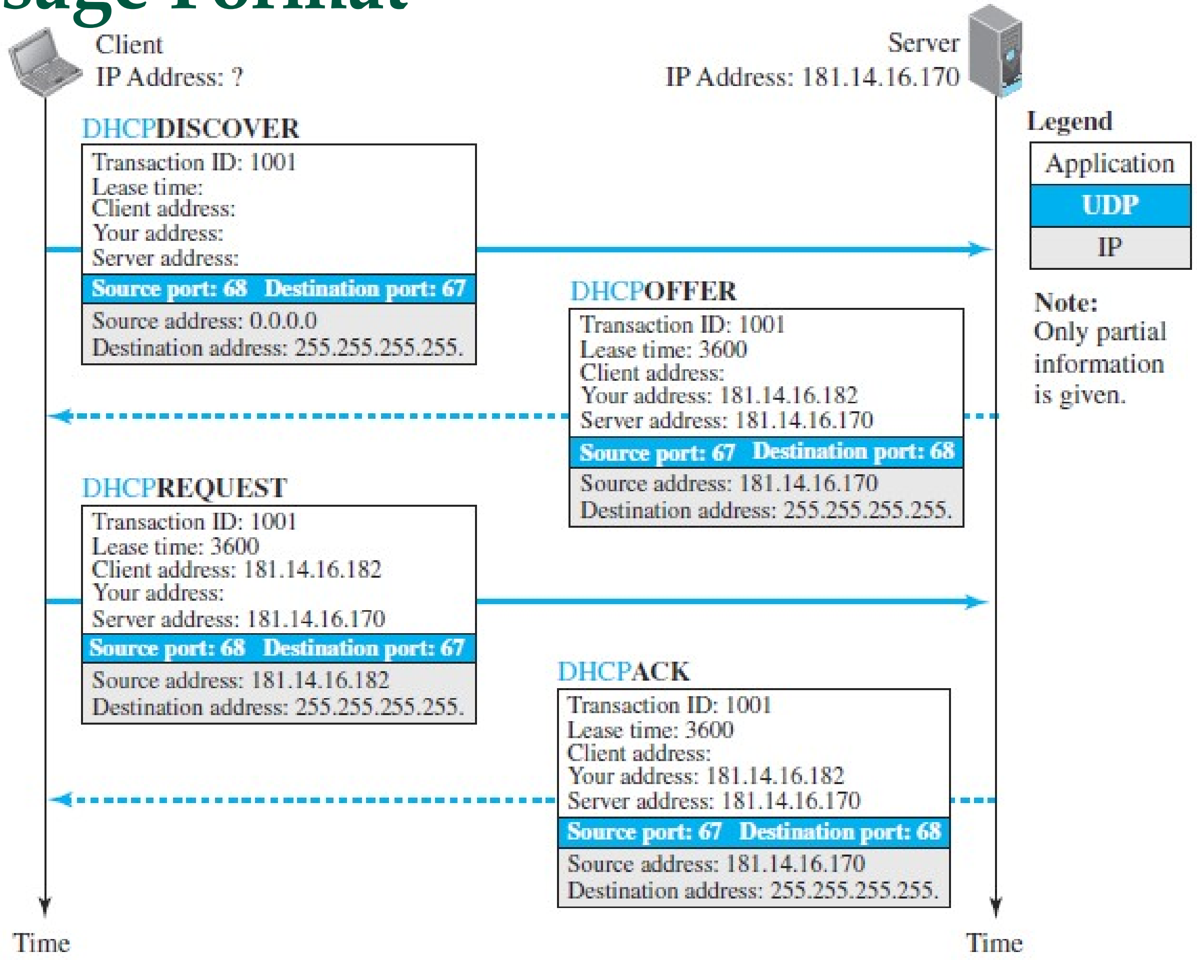
Opcode: Operation code, request (1) or reply (2)
Htype: Hardware type (Ethernet, ...)
HLen: Length of hardware address
HCount: Maximum number of hops the packet can travel
Transaction ID: An integer set by the client and repeated by the server
Time elapsed: The number of seconds since the client started to boot
Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used
Client IP address: Set to 0 if the client does not know it
Your IP address: The client IP address sent by the server
Server IP address: A broadcast IP address if client does not know it
Gateway IP address: The address of default router
Server name: A 64-byte domain name of the server
Boot file name: A 128-byte file name holding extra information
Options: A 64-byte field with dual purpose described in text

Option format

1 DHCPDISCOVER	5 DHCPACK
2 DHCPOFFER	6 DHCPNACK
3 DHCPREQUEST	7 DHCPRELEASE
4 DHCPDECLINE	8 DHCPINFORM



DHCP Message Format



References

- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.